

KOUSIK KUMAR DUTTA

Ph.D. in Computer Science & Engineering
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About

Ph.D. researcher in Computer Science and Engineering at IIT Ropar, specializing in spatial data management, constrained path query processing, and scalable graph algorithms for time-dependent networks. My research focuses on efficient algorithms and data-management frameworks for preference-aware routing, interval-based optimization, and flexible query processing in dynamic transportation networks.

Research Interests

- **Spatial Data Management:** Efficient storage, querying, and processing of large-scale spatio-temporal and road network data.
- **Scalable Graph Algorithms and Path Query Processing:** Efficient search, labeling, and constrained path-query methods for large dynamic and time-dependent networks.
- **Intelligent Transportation Systems:** Data-driven approaches for mobility analytics, navigation systems, and transportation optimization.

Education

Ph.D. in Computer Science and Engineering 2021–2026 *Indian Institute of Technology Ropar, Punjab, India*

- **Dissertation:** “Constrained Maximization of Preferences on Time-Dependent Road Networks”
- **Advisor:** Dr. Venkata M. V. Gunturi & Venkata Kalyan Tavva
- **CGPA:** 7.92/10.00

M.Tech in Computer Science and Engineering 2019–2021 *Indian Institute of Technology Ropar, Punjab, India*

- **Thesis:** “A Multi-Threading Algorithm for Constrained Path Optimization on Road Networks”
- **CGPA:** 8.39/10.00

B.Tech in Computer Science and Engineering 2014–2018 *Kalyani Government Engineering College, West Bengal, India*

- **CGPA:** 7.45/10.00

Publications

Journal Publications

- [J1] **Kousik Kumar Dutta**, Venkata M. V. Gunturi. A Parallelizable Algorithm for Constrained Maximization of Preferences in Large Time-Dependent Graphs. *Geoinformatica*, vol. 30, article 19, 2026. DOI: <https://doi.org/10.1007/s10707-026-00577-z>

Conference Publications

- [C1] **Kousik Kumar Dutta**, Venkata M. V. Gunturi. A User-Configurable Navigation System for Wideness and Turn-Aware Routing. *International Conference on Mobile Data Management (MDM)*, 2026. DOI: <https://doi.org/10.1109/MDM71479.2026.00050>
- [C2] **Kousik Kumar Dutta**, Venkata M. V. Gunturi. Interval-Based Constrained Path Optimization in Time-Dependent Road Networks. *26th International Conference on Web Information Systems Engineering (WISE)*, 2025. DOI: https://doi.org/10.1007/978-981-95-7251-9_12
- [C3] **Kousik Kumar Dutta**, Ankita Dewan, Venkata M. V. Gunturi. A Multi-Threading Algorithm for Constrained Path Optimization on Road Networks. *23rd International Conference on Web Information Systems Engineering (WISE)*, 2022. DOI: https://doi.org/10.1007/978-3-031-20891-1_9
- [C4] **Kousik Kumar Dutta**, Prathamesh N. Tanksale, Shirshendu Das. A Fairness Conscious Cache Replacement Policy for Last Level Cache. *Design, Automation and Test in Europe (DATE)*, 2021. DOI: <https://doi.org/10.23919/DATE51398.2021.9474096>

- [C5] Ankita Dewan, Venkata M. V. Gunturi, Vinayak Naik, **Kousik Kumar Dutta**. NEAT Activity Detection Using Smartwatch at Low Sampling Frequency. *IEEE SmartWorld/SCALCOM/UIC/ATC/IOP/SCI*, 2021. DOI: <https://doi.org/10.1109/SWC50871.2021.00014>

Workshop Publications

- [W1] **Kousik Kumar Dutta**, Venkata M. V. Gunturi. Constrained Path Optimization on Time-Dependent Road Networks. *WISE PhD Symposium, Demos and Workshops*, 2024. DOI: https://doi.org/10.1007/978-981-96-1483-7_24
- [W2] Prathamesh N. Tanksale, **Kousik Kumar Dutta**, Shirshendu Das, T. V. Kalyan. CAMOUFLAGE: An Efficient Mechanism to Hide Congestion in NVM LLC. Extended Abstract, *IEEE International Conference on High Performance Computing (HiPC)*, 2023. DOI: <https://doi.org/10.1109/HiPCW61695.2023.00021>

Manuscripts Under Review

- [U1] **Kousik Kumar Dutta**, Venkata M. V. Gunturi. A Bi-Directional Search for Interval Constrained Path Optimization in Large Time-Dependent Road Networks. *Knowledge and Information Systems*, 2026. (Major revision submitted)
- [U2] **Kousik Kumar Dutta**, Venkata M. V. Gunturi. A Decomposition-Based Heuristic for Selective Pickup–Delivery Routing with Non-Additive Loading–Unloading Costs. *Annals of Operations Research*, 2026. (Under review)
- [U3] **Kousik Kumar Dutta**, Venkata M. V. Gunturi. PACE: An Efficient Framework for Constrained Preference Path Profiles in Large Time-Dependent Road Networks. *Information Systems*, 2026. (Under review)

Teaching Experience

Teaching Assistant, Indian Institute of Technology Ropar

- **Databases (B.Tech. Core Course)** 7 Offerings (2021–2025)
 - Conducted laboratory sessions and evaluated assignments on SQL, relational database design, and query processing.
 - Mentored student teams in the development of database-driven applications and course projects.
- **Software Engineering (B.Tech. Core Course)** 5 Offerings (2020–2025)
 - Evaluated laboratory assignments and guided students through software development practices.
 - Assisted student teams in requirement analysis, software design, testing, and project implementation.
- **PG Software Lab (M.Tech. Core Course)** 1 Offering (2020)
 - Mentored postgraduate students in advanced data structures and algorithm implementation.

Academic Services

- **Reviewer** for *Geoinformatica* (2023–Present); completed peer review of three manuscripts in areas including spatial databases, graph processing, and route planning.
- **Technical Committee Head**, *SYNTACS 2026* (Symposium on Novel Technologies and Advances in Computer Science), Department of Computer Science and Engineering, Indian Institute of Technology Ropar.

Awards, Fellowships, and Grants

- **IEEE TCDE Student Grant (2026)**: Recipient of the IEEE Technical Committee on Data Engineering (TCDE) Student Grant, awarded competitively to support participation in the International Conference on Mobile Data Management (MDM 2026).
- **Institute Fellowship (Ph.D.)**: Awarded full institute fellowship for doctoral studies at Indian Institute of Technology Ropar (2021–2026).
- **MHRD Scholarship (M.Tech.)**: Awarded Ministry of Human Resource Development (MHRD) scholarship during M.Tech studies at Indian Institute of Technology Ropar (2019–2021).
- **GATE 2019**: Secured All India Rank (AIR) 644 in Computer Science and Engineering.

References

Available upon request.